

Global Regularity Framework for Arbitrary Initial Data in L^2

1 Introduction

The goal of this framework is to establish global regularity for the Navier–Stokes equations without modifying the equations or imposing restrictive assumptions on the initial data. Specifically, we aim to prove regularity for arbitrary initial conditions $u_0 \in L^2(\mathbb{R}^3)$. This approach represents a departure from traditional methods that rely on higher Sobolev space assumptions, such as H^1 , by utilizing principles that hold universally.

1.1 Key Principles

This framework integrates the following key principles:

1. ****Scale-Barrier Principle:**** Controls high-frequency energy transfer through spectral decay, preventing blowup at small scales.
2. ****Monotone Functional:**** Tracks global energy and enstrophy through the boundedness of $Q(t)$.
3. ****Non-Sobolev Compactness:**** Establishes strong convergence in L^2 without relying on classical Sobolev embeddings.
4. ****Curvature-Based Regularity:**** Regulates velocity gradients through geometric arguments involving Ricci curvature.

2 Initial Conditions and Universality

We begin with the most general assumption for initial data:

$$u_0 \in L^2(\mathbb{R}^3), \quad \|u_0\|_{L^2} < \infty. \quad (1)$$

This ensures finite initial kinetic energy while avoiding unnecessary constraints. The key challenge is to propagate this minimal regularity forward in time to establish smooth solutions for all $t \geq 0$.

2.1 Challenges with Arbitrary Initial Data

Traditional approaches often require $u_0 \in H^1$ or higher Sobolev spaces to:

- Ensure finite enstrophy for controlling velocity gradients.

- Derive strong compactness results using Sobolev embeddings.
- Simplify energy propagation and spectral decay.

This framework overcomes these challenges by relying on spectral decay and compactness arguments that hold directly in L^2 .

2.2 Framework for Generality

The proof proceeds through the following steps:

1. **Energy Boundedness:** Ensure the global boundedness of $Q(t)$ using:

$$Q(t) = \|u(t)\|_{L^2}^2 + \|\nabla u(t)\|_{L^2}^2. \quad (2)$$

2. **Spectral Decay:** Utilize the Scale-Barrier Principle to enforce:

$$E(k, t) \leq Ck^{-\beta}, \quad \beta > 3, \quad (3)$$

ensuring suppression of high-frequency energy contributions.

3. **Compactness:** Achieve strong convergence in L^2 through Non-Sobolev Compactness by controlling high-frequency tails and leveraging bounded energy.
4. **Gradient Control:** Use curvature-based arguments to bound $|\nabla u|^2$, ensuring velocity gradients remain finite for all time.

3 Proof Overview

The proof integrates the four principles to establish smooth solutions:

1. Start with $u_0 \in L^2$ and propagate the boundedness of $Q(t)$ using Gronwall's inequality and dissipation dominance.
2. Apply the Scale-Barrier Principle to enforce spectral decay, ensuring energy does not cascade indefinitely to high wave numbers.
3. Use compactness arguments to achieve strong convergence in L^2 for approximate solutions.
4. Employ geometric arguments to bound velocity gradients, completing the proof of smoothness for all $t \geq 0$.

4 Conclusion

This framework provides a robust and general approach to proving global regularity for the Navier–Stokes equations. By avoiding restrictive assumptions on initial data and relying on principles that hold universally, it achieves a more ambitious goal while maintaining mathematical rigor.

References

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